ANALAB INJECTION

COMPOSITION

Each ml contains : **Tramadol Hydrochloride** **50 mg**

DOSAGE FORM AVAILABLE

Solution for injection

PRODUCT DESCRIPTION

Clear and colorless solution filled in colorless ampoules.

ACTIONS

Tramadol is a centrally acting opioid analgesic. It is a nonselective pure agonist at μ , δ and κ opioid receptors with a higher affinity for the μ receptor. Other mechanisms, which contribute to its analgesic effect, are inhibition of neuronal re-uptake of noradrenaline and enhancement of serotonin release. Tramadol has an antitussive effect. In contrast to morphine, analgesic doses of tramadol over a wide range have no respiratory depressant effect.
Also, gastrointestinal motility is not affected. Effects on the cardiovascular system tend to be slight. The potency of tramadol is reported to be 1/10-1/6 that of morphine. The onset of the analgesic effect of Tramadol hydrochloride is rapid and the effect lasts for several hours. Tramadol hydrochloride is a potent drug and should not be used for minor pain. It does not have an anti-inflammatory effect. Tramadol hydrochloride has an analgesic effect in all the usual test models for centrally acting analgesics. On parenteral administration, the intensity of the effect is equivalent to that of other opioid analgesic, e.g. pentazocine and pethidine. On oral administration, Tramadol hydrochloride is practically as effective as on parenteral application.
The onset of activity is rapid and the duration of the effect is similar to that morphine. Tramadol hydrochloride is a pure opiate agonist. The morphine antagonist, naloxone, neutralizes the analgesic effect of Tramadol hydrochloride.

Pharmacokinetics : Serum protein-binding is low (about 20%). The elimination half-life in the terminal phase is 6 hours with about 1/4-1/3 of the active substance appearing unchanged in the urine. One of the metabolites is pharmacologically active, but the concentration in the blood is lower than that of Tramadol hydrochloride itself.

Excretion : Tramadol hydrochloride and its metabolites are almost completely eliminated via the kidneys.

INDICATIONS

Moderate to severe pain.

DOSAGE AND ADMINISTRATIONS

The dosage depends on the intensity of the pain and the sensitivity of the individual patient. ANALAB INJECTION should be administered as follows :

Single dose :

For adults and adolescents (12 years and older) :
ANALAB INJECTION is not approved for use in patients below 12 years old.
I.V. 1-2 ampoules-injected slowly or diluted in solution for injection and infused
I.M. 1-2 ampoules
S.C. 1-2 ampoules
If satisfactory pain relief is not obtained within 30-60 minutes of the administration of 50 mg Tramadol hydrochloride, a 2nd dose of 50 mg may be given.
If clinical experience has shown that a higher dose (2 ampoules) is probably necessary in cases of severe pain, the higher dose may be used for the initial dose.
In general, a daily dosage of 400 mg should not be exceeded.
For the treatment of severe post-operative pain, the higher doses in the first few hours may be given.
Generally, over 24 hours more than the normal doses are not necessary.

- Paediatric population :

The safety and efficacy of ANALAB INJECTION has not been studied in the paediatric population. Therefore, use of ANALAB INJECTION is not recommended in patients under 12 years of age.

Geriatic patients : In acute pain a dosage adjustment is not necessary. In chronic pain the dosage intervals may have to be extended.
because elimination may be prolonged in old patients (usually >75 years) with no clinically manifest hepatic or renal insufficiency.

Hepatic and Renal Insufficiency/Dialysis : In acute pain a dosage adjustment is not necessary. In patients with severe renal and/or hepatic insufficiency ANALAB INJECTION should not be administered.
In less severe cases the dosage interval should be prolonged according to the patient's requirements.

Duration of Treatment : ANALAB INJECTION should not be given for longer than necessary. Patients with a tendency of drug abuse or dependence should only be given ANALAB INJECTION for short periods and under strict medical supervision.

OVERDOSAGE

Symptoms : The cardinal signs of a tramadol overdose are mydriasis or miosis and respiratory depression up to respiratory failure.
Other typical symptoms include clouding of consciousness to the point of coma, generallized epileptic seizures, hypotension, tachycardia, vomiting, cardiovascular collapse.

Treatment : The signs and symptoms of tramadol intoxication can be abolished by administration of an opioid antagonist (e.g., naloxone).
which should be administered carefully in repeated small doses, since its duration of action is shorter than that of tramadol, aspiration and emptying stomach. However, naloxone administration for tramadol overdose may increase the risk of seizures.
In addition, intensive care measures (intubation and mechanical ventilation in particular) should be instituted. The use of benzodiazepines should be considered for patients with seizures.
Measures aimed at protecting against the loss of heat as well as volume replacement therapy may also become necessary.
Hemodialysis or hemofiltration alone is not sufficient or suitable due to the slow elimination of tramadol from the serum by these route.

CONTRAINDICATIONS

Known hypersensitivity to tramadol, acute intoxication with alcohol, hypnotics, analgesics or other CNS-acting drug. Patients who are receiving MAO inhibitors or who have taken them within the last 14 days.

- Children younger than 18 years to treat pain after surgery to remove the tonsils and/or adenoids.
- Adolescents between 12 and 18 years who are obese or have conditions such as obstructive sleep apnea or severe lung disease, which may increase the risk of serious breathing problems.

PRECAUTION AND WARNINGS

- Dependence on opioids :
  Tramadol is not recommended as a substitute in opioid-dependent patients. Although tramadol is an opiate agonist, it cannot suppress opioid i.e., morphine, withdrawal symptoms. Because of the difficulty in assessing dependence in patients who have previously.
  received substantial amounts of opioid agonists, tramadol should be used with caution in patients with such a patients.
- Reduced level of consciousness disorders of uncertain origin.
- Patients in shock should be carefully monitored during treatment
- Respiratory depression :
  Administer ANALAB INJECTION cautiously in patients at risk for respiratory depression, including patients with substantially decreased respiratory reserve, hypoxia, hypercapnia, or pre-existing respiratory depression,as in these patients,even therapeutic doses of ANALAB INJECTION may decrease respiratory drive to the point of apnea. In these patients, alternative non-opioid analgesics should be considered. When large doses of tramadol are administered with anaesthetic medication or alcohol, respiratory depression may result. Respiratory depression should be treated as an overdose. If naloxone is to be administered, use cautiously because it may precipitate seizures.
- Increase in intracranial pressure due to head injuries or brain diseases tramadol should be used with caution in patients with increased intracranial pressure due to head injuries or brain diseases, since the respiratory depressant effects of opiate agonists include carbon dioxide retention and secondary elevation of cerebrospinal fluid pressure and such effects may be markedly exaggerated in these patients. Also, pupillary changes (miosis) from tramadol may obscure the existence, extent or course of intracranial pathology. Clinicians also should maintain a high index of suspicion for adverse drug reaction when evaluating alter mental status in this patients if they are receiving tramadol.
- In patients sensitive to opiates the medicinal product should only be used with caution
- Patients with tendency to drug abuse.
  Tramadol has a low dependence potential. On long-term use tolerance, psychotic and physical dependence may develop. In patients with a tendency to drug abuse or dependence, treatment should only be carried out for short periods under strict medical supervision.
- Paediatric population : The safety and efficacy of ANALAB INJECTION has not been studied in the paediatric population. Therfore, use of ANALAB INJECTION is not recommended in patients under 12 years of age.
- To avoid alcohol.
  Tramadol should not be used concomitantly with alcohol consumption. The use of tramadol in patients with liver disease is not recommended.
- Cytochromes P450 (CYP) 2D6 Ultra-Rapid Metabolism : Some individuals may be CYP2D6 ultra-rapid metabolizers. These individuals convert tramadol more rapidly than other people into its more potent opioid metabolites O-desmethyltramadol (M1).
  This rapid conversion could result in higher than expected opioid-like side effects including life-threatening respiratory depression.
  The prevalence of this CYP2D6 phenotype varies widely and has been estimated at 0.5 to 1% in Chinese, Japanese and Hispanics, 1 to 10% in Caucasians, 3% in African, Americans, and 16-28% in North Africans, Ethiopians, and Arabs. Data are not available for other ethnic groups.
- Risks from Concomitant Use with Benzodiazepines :
  Profound sedation, respiratory depression, coma, and death may result from the concomitant use of ANALAB INJECTION with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

FRONT

  If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.
  If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.
  If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.
  Follow patients closely for signs and symptoms of respiration depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when ANALAB INJECTION is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (see DRUG INTERACTIONS).

- Serotonin Syndrome with Concomitant Use of Serotonergic Drugs:

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of ANALAB INJECTION with serotonergic drugs (See DRUG INTERACTIONS). This may occur within the recommended dosage range.
Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See DRUG INTERACTIONS). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue ANALAB INJECTION if serotonin syndrome is suspected.

- Adrenal Insufficiency:

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure.
If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency.
The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

- Sexual Function/Reproduction:

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)
Use in pregnancy : Tramadol has been shown to cross the placenta. There are no adequate and well-controlled studies in pregnant women. Safe use in pregnancy has not been established. ANALAB INJECTION is not recommended for pregnant women.
Use in lactation : Approximately 0.1% of the maternal dose of tramadol is excreted in breast milk. In the immediate post-partum period, for maternal oral daily dosage up to 400 mg, this corresponds to a mean amount of tramadol ingested by breast-fed infants of 3% of the maternal weight-adjusted dosage. For this reason tramadol should not be used during lactation or alternatively, breast-feeding should be discontinued during treatment with tramadol. Discontinuation of breast-feeding is generally not necessary following a single dose of tramadol.

SIDE EFFECTS

There have been occasional reports of nausea, sweating, dry mouth, dizziness, lightheadedness, headache, motorial weakness, changes in appetite, blurred vision, micturition disorders, worsening asthma, epileptiform convulsion, increase intracranial pressure and bradycardia, dependence, increase liver enzymes, affect reactions.
In very rare cases, various psychic side effects may occur depending on personality and duration of treatment. These include changes in mood, changes in activity and changes in cognitive and sensorial capacity (e.g. decision behavior, perception disorder).
Allergic reactions (e.g. dyspnoea, bronchospasm, wheezing, angioneurotic oedema) and anaphylaxis have also been reported.
In rare cases, there may be effects on cardiovascular regulation (palpitation, tachycardia, postural hypotension up to cardiovascular collapse). These side effects may occur particularly on IV administration and in patients under physical strain. Moreover, headache, retching, vomiting, constipation, gastrointestinal irritation (e.g. feeling of pressure in the stomach, bloating) and dermal reactions (e.g. pruritus, exanthema) may occur.
Very rarely epileptiform convulsions have been reported. They occurred after IV administration of high doses of tramadol or concomitant administration with drugs which can lower the seizure threshold or themselves induce cerebral convulsions (e.g. antidepressants or neuroleptics).
Respiratory depression (rare) has been reported. However, if the recommended doses are considerably exceeded or centrally depressant drugs are administered concomitantly, respiratory depression may occur.

Postmarketing Experience:

Serotonin syndrome (See PRECAUTIONS AND WARNINGS)
Adrenal insufficiency (See PRECAUTIONS AND WARNINGS)
Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility.
The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.
Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

DRUG INTERACTIONS

Concurrent use of tramadol and other CNS-depressant drugs and/or alcohol may enhance the central adverse effects of tramadol. respiratory depression in particular. Concomitant treatment with tramadol and antipsychotic drugs has occasionally been associated with epileptic seizures.
Carbamazepine markedly reduces the serum concentrations of tramadol and may reduce both the analgesic effect and the duration of action of tramadol.
Patients treated with MAOIs within 14 days prior to administration of the opioid pethidine have experienced life-threatening interactions affecting the central nervous system as well as the respiratory and circulatory centers. Similar interactions with MAOIs cannot be ruled out for tramadol either.
Concomitant administration of tramadol with cimetidine (enzyme inhibitor) does not result in clinically significant changes in tramadol pharmacokinetics. Therefore, no alteration of tramadol dosage regimen is recommended.
Concomitant use of tramadol with mixed agonist/antagonists (e.g., buprenorphine, nalbuphine, pentazocine) is not recommended because the analgesic effect of a pure agonist may be reduced.
Tramadol increases the seizure risk in patients taking selective serotonin-reuptake inhibitors (SSRIs), tricyclic antidepressants, neuroleptics and other seizure threshold lowering drugs.
Other drug known to inhibit the CYP3A4 isoenzyme of cytochrome P450, such as ketoconazole and erythromycin may inhibit the metabolism (via N-demethylation) of tramadol and probably the metabolism of M1. The clinical importance of such an interaction has not been established.
There is a theoretical possibility that tramadol could interact with lithium due to their respective mechanisms of action. ANALAB INJECTION is incompatible with injection solutions of diclofenac, indomethacin, phenylbutazone, diazepam, flunitrazepam, midazolam and nitroglycerin.

- Benzodiazepines :
  Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.
  The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABA$_A$ sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.
  Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see PRECAUTIONS AND WARNINGS).
  Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.
- Serotonergic Drugs:

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue ANALAB INJECTION if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See PRECAUTIONS AND WARNINGS).

STORAGE

Store at a temperature not exceeding 30^{o}C and protect from light.
Keep medicines out of reach of children.
Jauhkan daripada kanak-kanak.
Shelf-life : 3 years.

SUPPLY

Printing box of 10x1 ml ampoules.

Date of revision : December 2017

BIOLAB
Manufactured by:
BIOLAB CO., LTD.
625 Bangpoo Industrial Estate,
Samutprakarn, Thailand

EEI039

BACK

Code : EEI039
Draft Revision : 10
Final Artwork Revision : A
Name : AW_LEAFLET ANALAB INJ MAL.ai
Remark :

Design by	Review by	Approved by
Signature	Signature	Signature
Date	Date	Date

Specification

Size : 140 x 240 MM. (+/- 3 MM.)
Color : พิมพ์ 1 สี คือ สีดำ
Material : ใช้กระดาษปอนด์ น้ำหนัก 55 แกรม
Other : พิมพ์ด้านหน้า หลัง บนแผ่นเดียวกัน

F-QCP-GE-011

Note :

#0=ทำ A/W ให้ RA
#1=เปลี่ยนชื่อเป็น "ANALAB INJECTION"
#2=แก้ไขคำผิด
#3=เพิ่มข้อความ"Precaution, Drug Interaction"
#4=อก้ไขคำผิด
#5=แก้ไขคำผิด
#6=แก้ไขคำผิด
#7=ลบหัวข้อ"TOXICOLOGY" ออก
#8=แก้ไขเดือน
#9=เพิ่มข้อความหัวข้อ"Precaution, side effects, drug interaction"
#10=แก้ไขคำผิด

○ ลูกค้าพิจารณาอนุมัติ ○ ทำเสนอราคา ○ สั่งพิมพ์
☑ RA ขึ้นทะเบียน ○ อื่นๆ..

COLOR GUIDE

BLACK 100%

BIOLAB